Title: Digital Inventory Tracking for Small Shops

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ABSTRACT

The Digital Inventory Tracking for Shops project is designed to help small retail shop owners maintain accurate records of their products and stock levels. In many small shops, inventory is still managed manually, which can lead to errors, misplaced records, and difficulty in tracking available items. This project provides a digital solution where product details, quantities, and stock updates can be recorded and stored in an organized manner.

The application allows shopkeepers to add new items, update stock quantities, and monitor product availability easily. It also helps identify low-stock items so that they can be restocked on time. By maintaining inventory digitally, the project reduces manual effort, improves accuracy, and helps shop owners manage their shop stock more efficiently.

EXISTING SYSTEM

In many small shops, inventory is maintained using manual methods such as notebooks, paper records, or basic spreadsheets. Shopkeepers record product details, stock quantities, and sales by hand. This method is time-consuming and often leads to mistakes such as incorrect entries, lost records, or difficulty in finding specific information when needed.

The manual approach also makes it hard to track stock levels accurately and identify items that need restocking. As the number of products increases, managing inventory becomes more complicated and inefficient. Because of these limitations, shop owners may face problems like stock shortages, overstocking, and delays in checking product availability.

PROPOSED SYSTEM

PThe proposed system introduces a digital inventory tracking solution designed for small shops to manage their stock records efficiently. In this system, shopkeepers can store product details such as item name, quantity, price, and category in a digital database. The application allows users to add new products, update stock information, and view available inventory quickly and accurately.

This system helps reduce manual errors and makes inventory handling faster and more organized. It also allows shop owners to monitor stock levels and identify low-quantity items that need restocking. By maintaining records digitally, the proposed solution improves accuracy, saves time, and helps shopkeepers maintain better control over their inventory.

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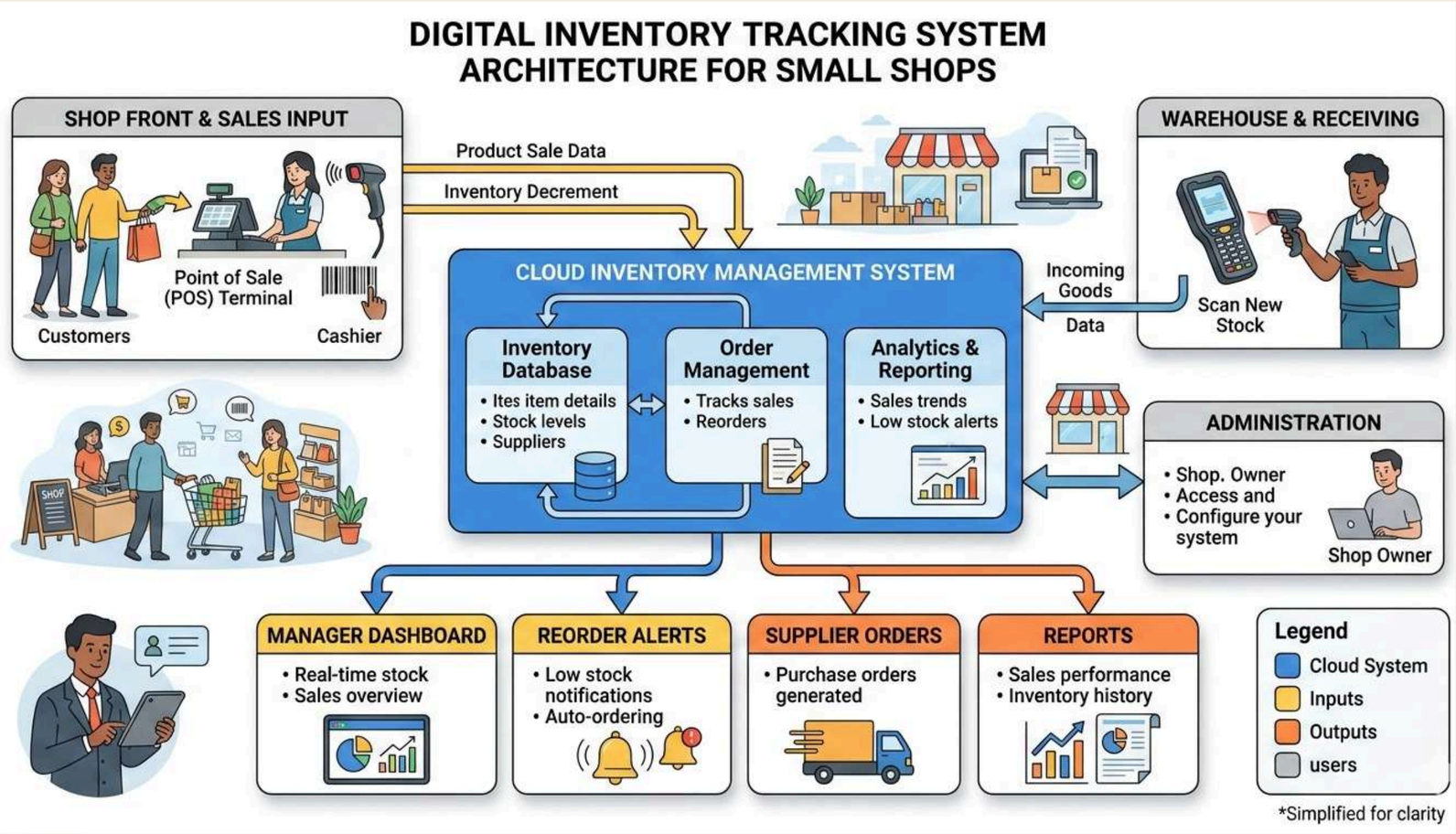
SYSTEM ARCHITECTURE



PROJECT MODULES

1. User Authentication Module

- Allows shopkeepers to securely log in to the system.
- Ensures that only authorized users can access and modify inventory data.

2. Product Management Module

- Add new products with details like name, category, quantity, and price.
- Edit or delete existing product information.

3. Inventory Update Module

- Update stock levels when new products arrive or items are sold.
- Track low-stock items and generate alerts for restocking.

4. Inventory Display Module

- View all available products along with quantities and prices.
- Search for specific items easily.

5. Reports Module

- Generate daily, weekly, or monthly inventory reports.
- Helps in analyzing stock trends and managing shop operations efficiently.

OBJECTIVES

1. Efficient Stock Management – To help shopkeepers maintain accurate records of all products and stock levels, reducing errors caused by manual tracking.

2. Time-Saving – To minimize the time spent on recording, updating, and checking inventory by providing a digital solution.

3. Low-Stock Alerts – To notify shop owners when products reach low stock levels, ensuring timely restocking and avoiding shortages.

4. Easy Product Tracking – To enable shopkeepers to search and track products quickly, making inventory monitoring more organized and convenient.

5. Reporting and Analysis – To generate inventory reports that help in understanding sales trends, stock movement, and overall shop performance.

6. Improved Accuracy – To reduce manual errors in stock calculations and ensure reliable inventory records.

CO'S ATTAINMENTS-DSA

CO1 : Algorithm Analysis, Searching and Sorting which involves searching and sorting algorithms such as linear search and bubble sort

CO2 : Abstract Data Types (Lists) which focuses on implementing the List ADT using a linked list to store product details

CO3 : Stacks and Queues which applies stacks and queues for operations like undo actions and order processing

CO4 : Hashing which uses hashing through a HashMap to enable fast product lookup and efficient data management.

CO'S ATTAINMENTS-FWD

CO1: Design the structure of a web page using HTML elements such as forms, tables, inputs, and buttons.

CO2: Apply CSS styling to improve the appearance and layout of the web application.

CO3: Implement JavaScript functions to perform dynamic operations like adding product details and deleting rows from the inventory table.

CO4: Develop a simple web-based inventory management system that helps store and manage product information for small shops.

SCREEN-SHOTS OF FWD & DSA

```java
class CO1_SearchSort {

    static void bubbleSort(int arr[]) {
        for(int i=0;i<arr.length-1;i++){
            for(int j=0;j<arr.length-i-1;j++){
                if(arr[j] > arr[j+1]){
                    int temp = arr[j];
                    arr[j] = arr[j+1];
                    arr[j+1] = temp;
                }
            }
        }
    }

    static int linearSearch(int arr[], int key){
        for(int i=0;i<arr.length;i++){
            if(arr[i] == key)
```

```java
                return i;
        }
        return -1;
    }

    public static void main(String[] args) {

        int arr[] = {45,20,10,50,30};

        bubbleSort(arr);

        System.out.println("Sorted Array:");
        for(int i:arr)
            System.out.print(i+" ");

        int result = linearSearch(arr,30);

        System.out.println("\nElement found at index: "+result);
    }
}
```

```java
class CO2_LinkedList {

    static class Node{
        int data;
        Node next;

        Node(int data){
            this.data = data;
        }
    }

    Node head;

    void insert(int data){
        Node newNode = new Node(data);

        if(head == null)
            head = newNode;
        else{
            Node temp = head;
            while(temp.next != null)
                temp = temp.next;

            temp.next = newNode;
        }
    }

    void display(){
        Node temp = head;

        while(temp != null){
            System.out.print(temp.data+" ");
            temp = temp.next;
        }
    }

    public static void main(String[] args){

        CO2_LinkedList list = new CO2_LinkedList();
```

```java
        list.insert(10);
        list.insert(20);
        list.insert(30);

        System.out.println("Linked List:");
        list.display();
    }
}
```

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```java
import java.util.*;

class CO3_StackQueue {

    public static void main(String[] args){

        Stack<Integer> stack = new Stack<>();

        stack.push(10);
        stack.push(20);
        stack.push(30);

        System.out.println("Stack Pop: "+stack.pop());

        Queue<Integer> queue = new LinkedList<>();

        queue.add(1);
        queue.add(2);
        queue.add(3);

        System.out.println("Queue Remove: "+queue.remove());
    }
}
```

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```java
import java.util.*;

class CO4_Hashing {

    public static void main(String[] args){

        HashMap<Integer,String> inventory = new HashMap<>();

        inventory.put(101,"Rice");
        inventory.put(102,"Sugar");
        inventory.put(103,"Oil");

        System.out.println("Product with ID 102: "+inventory.get(102));

        System.out.println("Inventory List:");
        for(Integer key: inventory.keySet()){
            System.out.println(key+" -> "+inventory.get(key));
        }
    }
}
```

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Digital Inventory Tracking for Small Shops

Add Product

Product ID | Product Name | Quantity | Price

Add Product

Product ID	Product Name	Quantity	Price	Action

Digital Inventory Tracking for Small Shops

Add Product

Product ID | Product Name | Quantity | Price

Add Product

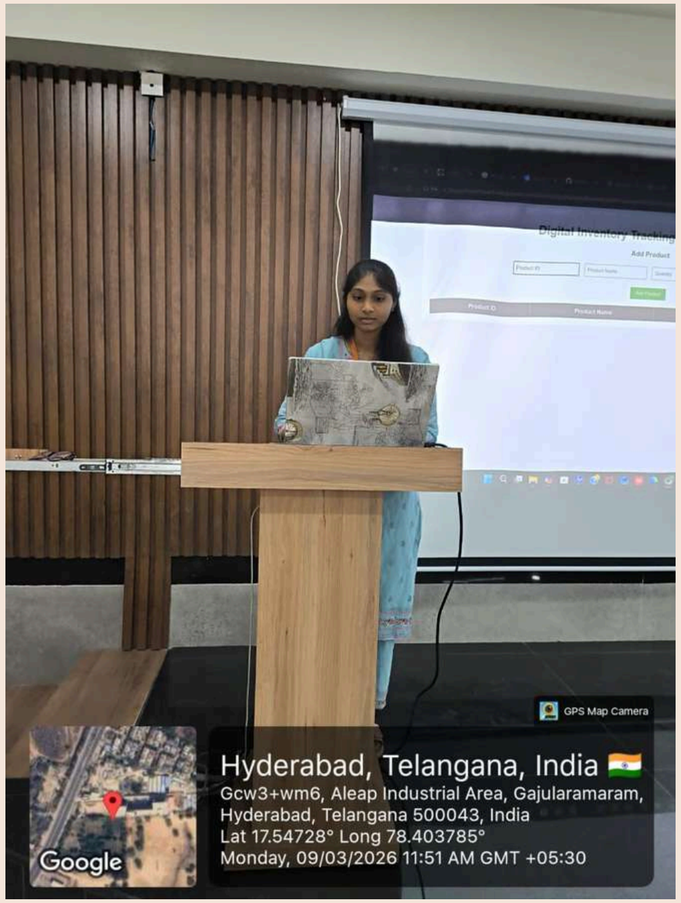
Product ID	Product Name	Quantity	Price	Action
1122	bottle	2	600	Delete

CONCLUSION

Digital Inventory Tracking for Small Shops provides an efficient way to manage products, stock levels, and sales using a computerized system instead of manual records. By using data structures such as lists, stacks, queues, hashing, and searching and sorting algorithms, the system can store and organize inventory data effectively. These techniques help shop owners quickly add, update, search, and sort product information, improving the overall management of the store.

Overall, the system helps reduce human errors, saves time, and ensures accurate tracking of available stock. It also improves decision-making by allowing shop owners to monitor product availability and sales easily. Implementing such a digital system makes inventory management faster, more reliable, and scalable for small businesses.

GEO-TAG


Digital Inventory Tracking
Add Product
GPS Map Camera
Google
Hyderabad, Telangana, India
Gcw3+wm6, Aleap Industrial Area, Gajularamaram,
Hyderabad, Telangana 500043, India
Lat 17.54728° Long 78.403785°
Monday, 09/03/2026 11:51 AM GMT +05:30

Thank you